



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

EXPERIMENT 8

Aim: Study of Mobile forensic tools.

1. Cellebrite UFED (Universal Forensic Extraction Device)

- **Overview:**

Cellebrite UFED is a comprehensive mobile forensic solution designed to extract, decode, and analyze data from mobile devices, SIM cards, memory cards, and cloud sources. It supports both logical and physical extractions, and is capable of bypassing device locks on many models.

- **Key Features:**

- Physical and logical extraction from iOS and Android devices.
- Bypass lock screens on certain devices.
- Cloud data acquisition capabilities.
- Data decoding and reporting for court-admissible evidence.

- **Primary Use:**

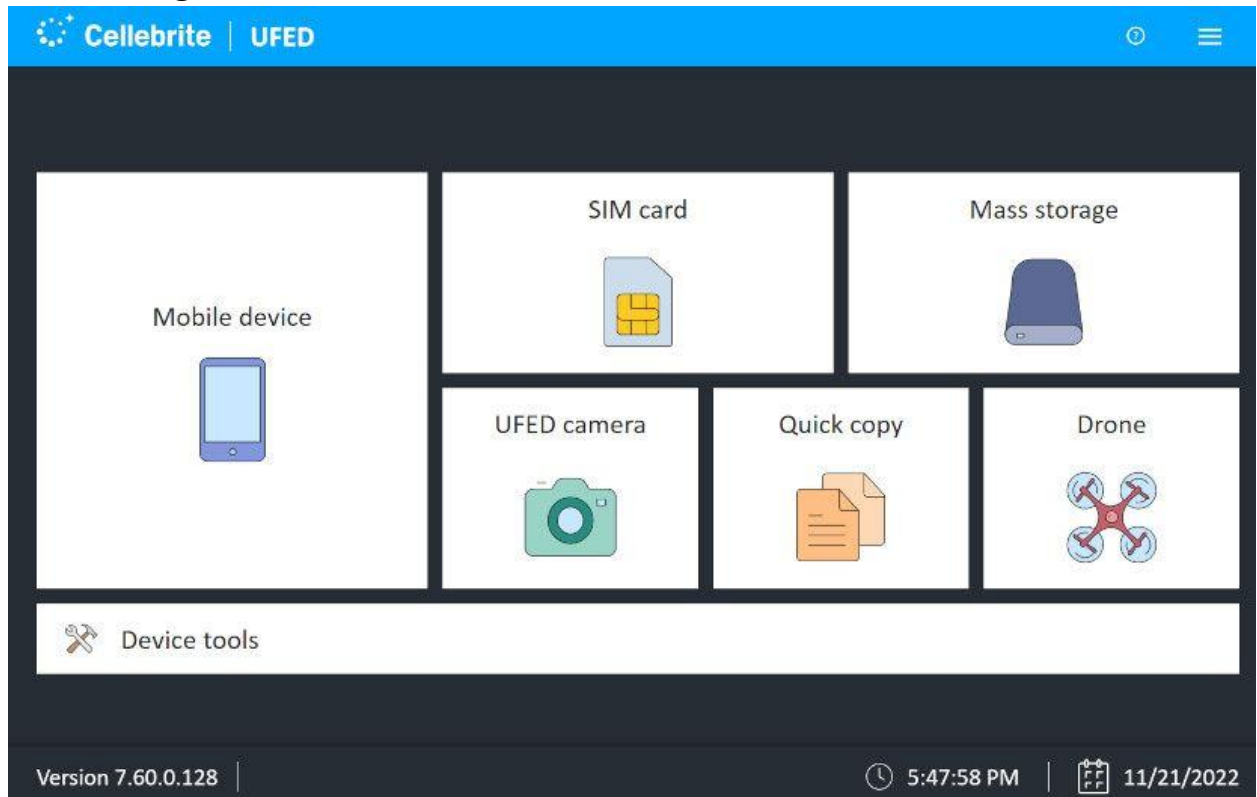
Frequently used by law enforcement, military, and intelligence agencies for criminal investigations and digital evidence collection.



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

- **Official Image:**



2. Magnet AXIOM

- **Overview:**

Magnet AXIOM is a digital forensics platform that recovers and analyzes evidence from mobile devices, computers, and cloud services. It organizes recovered data into timelines, connections, and artifacts for investigative purposes.

- **Key Features:**

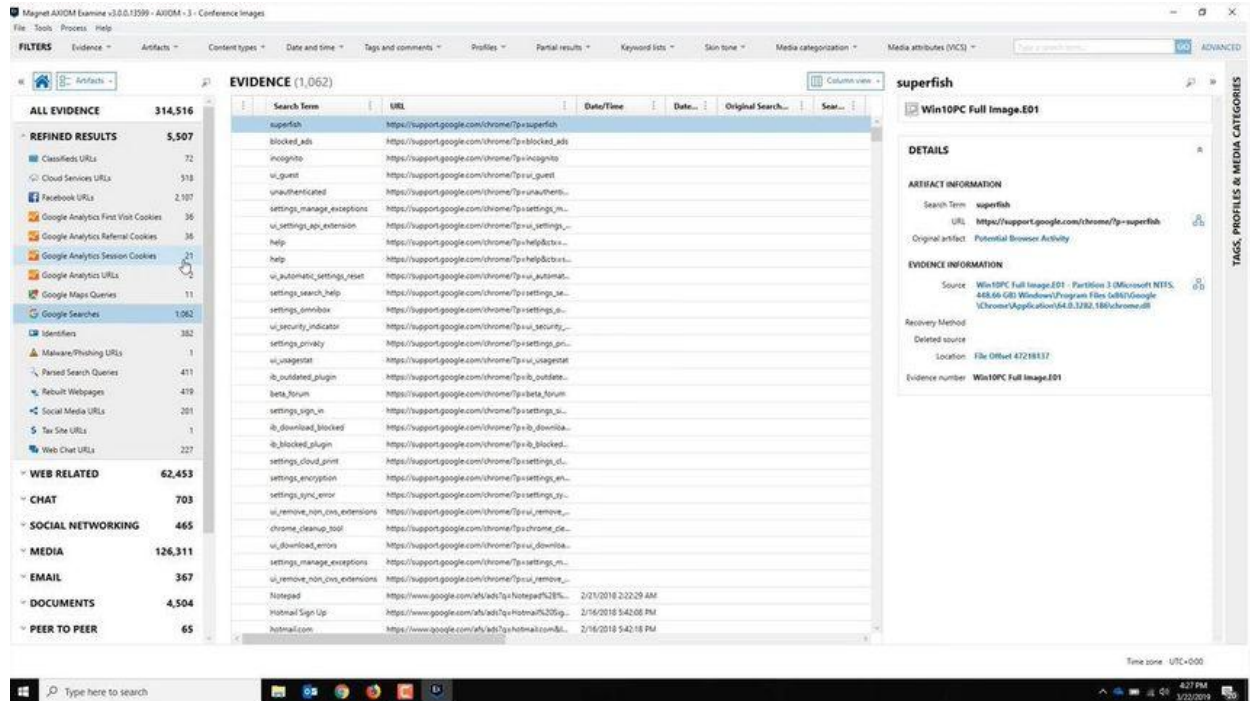
- Comprehensive mobile and cloud data recovery.
- Intuitive visualizations (timeline, connections, artifact explorer).
- Support for encrypted data recovery.
- Court-ready reporting tools.



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

- **Primary Use:**
Ideal for investigations requiring correlation of digital evidence across devices and platforms, including criminal cases, corporate investigations, and cybersecurity incidents.
- **Official Image:**



3. Oxygen Forensic Detective

- **Overview:**
Oxygen Forensic Detective is a comprehensive data extraction and analysis tool for mobile devices, cloud services, and IoT devices. It is particularly strong in password extraction, decryption, and recovery of hidden or deleted information.
- **Key Features:**
 - Data extraction from 40,000+ device profiles and apps.
 - Cloud data acquisition and account analysis.
 - Support for encrypted backups and password recovery.
 - Timeline, map, and link graph visualizations.



**SHRI VILEPARLE KELAVANI MANDAL'S
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING**
(Autonomous College Affiliated to the University of Mumbai)
NAAC ACCREDITED with "A" GRADE (CGPA : 3.18)



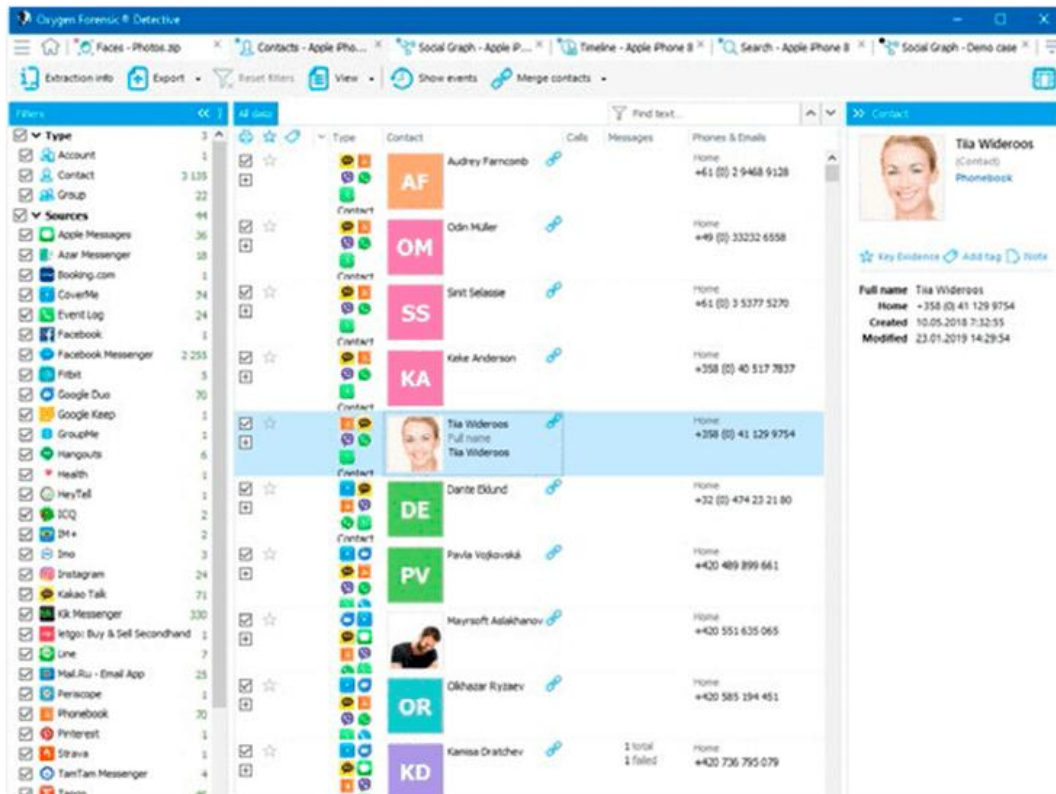
Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

- **Primary Use:**

Used for in-depth investigations where access to protected or cloud-stored data is critical, including corporate espionage, cybercrime, and complex criminal cases.

- **Official Image:**



Conclusion:

The study of three prominent mobile forensic tools — Cellebrite UFED, Magnet AXIOM, and Oxygen Forensic Detective — highlights the evolving sophistication and specialization within the field of mobile device forensics. Each tool offers distinct capabilities: Cellebrite excels at physical extraction and lock bypassing, Magnet AXIOM specializes in cross-platform evidence correlation and intuitive visualization, while Oxygen Forensic Detective provides in-depth cloud and IoT data extraction coupled with powerful decryption abilities.